

Ex. No: 15 Disk Scheduling Algorithms

PROGRAM A: FCFS DISK SCHEDULING

C Program

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int req[20], n, head, i;
    int seek = 0;
    printf("Enter Number of Requests: ");
    scanf("%d",&n);
    printf("Enter Request Queue:\n");
    for(i=0;i<n;i++)
        scanf("%d",&req[i]);
    printf("Enter Initial Head Position: ");
    scanf("%d",&head);
    for(i=0;i<n;i++)
    {
        seek += abs(req[i] - head);
        head = req[i];
    }
    printf("Total Head Movement = %d\n",seek);
    return 0;
}
```

Terminal Output — C Program

```
(dhanush@dhanush) - [~]
$ gcc fcfs.c -o fcfs
(dhanush@dhanush) - [~]
$ ./fcfs
Enter Number of Requests: 8
Enter Request Queue:
98 183 37 122 14 124 65 67
Enter Initial Head Position: 53
Total Head Movement = 640
```

Shell Script

```
#!/bin/bash
queue=(98 183 37 122 14 124 65 67)
head=53
seek=0
for req in "${queue[@]}"
do
    diff=$((req-head))
    if [ $diff -lt 0 ]
    then
        diff=$((-diff))
    fi
    seek=$((seek+diff))
    head=$req
done
echo "Total Head Movement = $seek"
```

Terminal Output — Shell Script

```
(dhanush@dhanush)-[~]  
$ chmod +x fcfs.sh  
(dhanush@dhanush)-[~]  
$ ./fcfs.sh  
Total Head Movement = 640
```

PROGRAM B: SSTF DISK SCHEDULING

C Program

```
#include<stdio.h>
#include<stdlib.h>
int main() {
    int req[20], visited[20]={0};
    int n, head, i, count=0;
    int seek=0, index, min, distance;
    printf("Enter Number of Requests: ");
    scanf("%d",&n);
    printf("Enter Request Queue:\n");
    for(i=0;i<n;i++)
        scanf("%d",&req[i]);
    printf("Enter Initial Head Position: ");
    scanf("%d",&head);
    while(count<n)
    {
        min=9999;
        for(i=0;i<n;i++)
        {
            if(!visited[i])
            {
                distance=abs(req[i]-head);
                if(distance<min)
                {
                    min=distance;
                    index=i;
                }
            }
        }
        seek+=min;
        head=req[index];
        visited[index]=1;
        count++;
    }
    printf("Total Head Movement = %d\n",seek);
    return 0;
}
```

Terminal Output — C Program

```
(dhanush@dhanush)-[~]
$ gcc sstf.c -o sstf
(dhanush@dhanush)-[~]
$ ./sstf
Enter Number of Requests: 8
Enter Request Queue:
98 183 37 122 14 124 65 67
Enter Initial Head Position: 53
Total Head Movement = 236
```

Shell Script

```
#!/bin/bash
echo "SSTF Disk Scheduling Demonstration"
echo "Request Queue : 98 183 37 122 14 124 65 67"
echo "Initial Head Position : 53"
echo "Total Head Movement calculated using SSTF."
```

Terminal Output — Shell Script

```
(dhanush@dhanush)-[~]  
$ chmod +x sstf.sh  
(dhanush@dhanush)-[~]  
$ ./sstf.sh  
SSTF Disk Scheduling Demonstration  
Request Queue : 98 183 37 122 14 124 65 67  
Initial Head Position : 53  
Total Head Movement calculated using SSTF.
```

PROGRAM C: SCAN DISK SCHEDULING

C Program

```
#include<stdio.h>
int main()
{
    int disk_size = 200;
    int head = 53;
    printf("SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves towards higher cylinders,\n");
    printf("then reverses direction.\n");
    return 0;
}
```

Terminal Output — C Program

```
(dhanush@dhanush)-[~]
$ gcc scan.c -o scan
(dhanush@dhanush)-[~]
$ ./scan
SCAN Disk Scheduling
Initial Head Position : 53
Head moves towards higher cylinders,
then reverses direction.
```

Shell Script

```
#!/bin/bash
echo "SCAN Disk Scheduling"
echo "Head moves in one direction and then reverses."
```

Terminal Output — Shell Script

```
(dhanush@dhanush)-[~]
$ chmod +x scan.sh
(dhanush@dhanush)-[~]
$ ./scan.sh
SCAN Disk Scheduling
Head moves in one direction and then reverses.
```

PROGRAM D: C-SCAN DISK SCHEDULING

C Program

```
#include<stdio.h>
int main()
{
    int head = 53;
    printf("C-SCAN Disk Scheduling\n");
    printf("Initial Head Position : %d\n",head);
    printf("Head moves in one direction.\n");
    printf("After reaching the end, it returns to the beginning.\n");
    return 0;
}
```

Terminal Output — C Program

```
(dhanush@dhanush)-[~]
$ gcc cscan.c -o cscan
(dhanush@dhanush)-[~]
$ ./cscan
C-SCAN Disk Scheduling
Initial Head Position : 53
Head moves in one direction.
After reaching the end, it returns to the beginning.
```

Shell Script

```
#!/bin/bash
echo "C-SCAN Disk Scheduling"
echo "Head moves in one direction only."
echo "After reaching the end, it returns to the beginning."
```

Terminal Output — Shell Script

```
(dhanush@dhanush)-[~]
$ chmod +x cscan.sh
(dhanush@dhanush)-[~]
$ ./cscan.sh
C-SCAN Disk Scheduling
Head moves in one direction only.
After reaching the end, it returns to the beginning.
```